Candidate name:

_____

Date: _____

Unofficial NUFYPET Math Simulation

Time: 1 hour (60 min.) Version 7.1
Total Marks: ____/30

Instructions

- Circle the answer you find right
- Calculators **MUST NOT** be used
- Figures are **NOT** accurately drawn and **NOT** to scale, except it is mentioned in a question
- There is only one answer for each question, so answers with two or more selected options marked as zero
- If you think something is wrong with the test, report a Methodist

Information

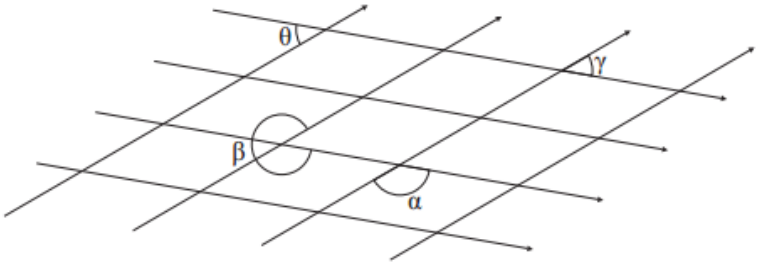
- The total mark for test is 30 (Each question marked as 1 point)
- There is **NO** penalty for wrong answers

Advice

- Read each question carefully
- Follow the time (There is an average 2 minutes for each question)
- Don't stay in one task for long
- Try to answer for all questions
- Recheck your answers if possible

Note: This is unofficial test made by MSDC

1. The diagram shows two sets of parallel lines. Angles θ, α, β and γ are as marked.



Which of the following statements are true?

1) $cos\,\alpha = -cos\,\theta$

2) $sin\,\beta = -sin\,\theta$

3) $tan\,\gamma = -tan\,\theta$

4) $sin\,(\theta + \alpha) = 1$

A 1 only **B** 3 only **C** 3 and 4 only **D** 1 and 2 only **E** 1, 2 and 4 only

2. Four individual spheres have radii:

$\frac{r}{2}$, r, $2r$ and $3r$

What is the sum of their surface areas?

A $58\pi r^2$ **B** $169\pi r^2$ **C** $57\pi r^2$ **D** $25\pi r^2$ **E** $26\pi r^2$

3. A computer game is on sale for €32.00.

The price ticket shows that this cost is a reduction of 20% of the original price.

What was the original price?

A €25.60 **B** €38.40 **C** €33.60 **D** €52.00 **E** €40.00

4. The United States Postal Service has an online guide to help people using their postal service. This is an extract:

First Class Mail

First class mail must be used for handwritten or personal correspondence.

First Class Mail Rates – single piece rates

First 30 grams Letter size mail	$0.44
Each additional 30 grams or part thereof	$0.17 up to 360 grams
First 30 grams Parcel	$1.22
Each additional 30 grams or part thereof	$0.17

Letters: 30 cm long, 16 cm wide, 1 cm thick, maximum weight 360 grams. (Any item larger than this must be sent as a parcel).

Parcels: Dimensions cannot exceed 1250 cm.

I want to send two items through the post using first class mail. One is 50 cm long, 25 cm wide, 10 cm thick and weighs 300 grams. The other is 20 cm long, 12 cm wide, 0.20 cm thick and weighs 110 grams.

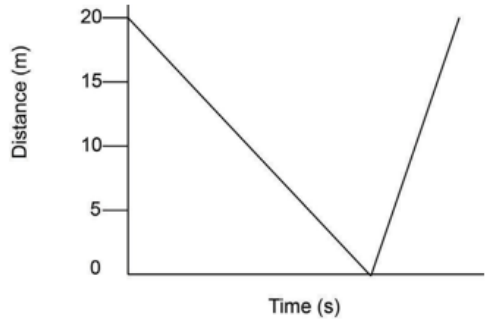
How much will it cost me to send the items?

A $2.92 **B** $3.53 **C** $3.70 **D** $4.04 **E** $4.48

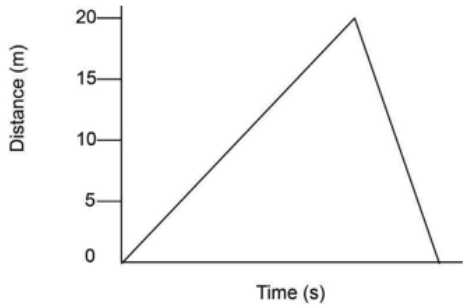
5. Sue takes her dog Freya to the park. She throws a stick for Freya to fetch. Freya runs to collect the stick and bring it back to Sue. While Freya is collecting the stick, Sue walks slowly towards Freya.

Which one of the following graphs could correctly show the distance between Sue and Freya?

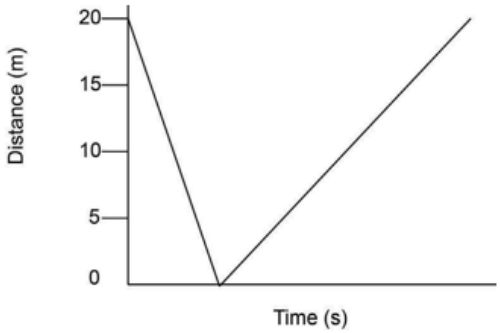
A



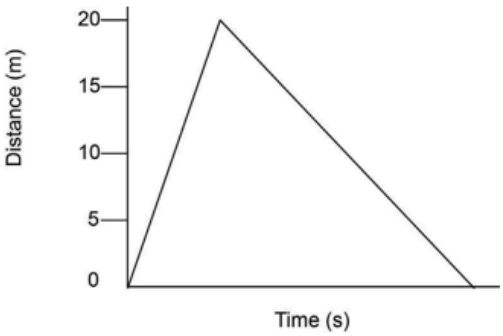
B



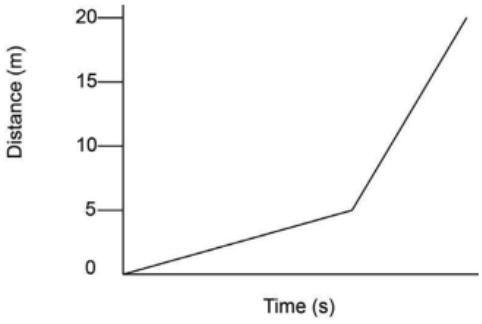
C



D



E



6. The variables x and y satisfy the following two equations:

$$x + 3y = 13$$

$$2x - y = 5$$

What is the value of $+y$?

A 7 **B** 8 **C** 6 **D** 9 **E** 10

7. The arithmetic mean of the three numbers a, b, c is 8.

Find the arithmetic mean of the four numbers: $a + 1$, $b + 2$, $c + 6$, 3.

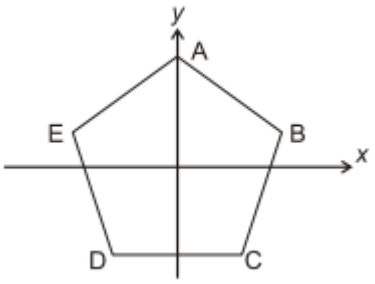
A 7 **B** 11 **C** 5 **D** 9 **E** 27

8. Evaluate:

$$(27^2 - 23^2) + (14^2 - 6^2)$$

A 1680 **B** 360 **C** 80 **D** 840 **E** 40

9. $ABCDE$ is a regular pentagon. The transformation R is a rotation about the origin and maps A to B, B to C, C to D, etc. The transformation S is a reflection in the y-axis.



Which of the following sequences of transformations (performed in the order that they are listed) would NOT leave vertex D in the same position?

A R R S R S R **B** S R **C** S R R S R S **D** R S R S **E** R S R R

10. A right-angled triangle has an area of $18\ cm^2$. One of the two shorter sides is twice the length of the other one.

What is the length of the hypotenuse of the triangle?

A $9\sqrt{2}$ cm **B** $6\sqrt{5}$ cm **C** $3\sqrt{10}$ cm **D** $3\sqrt{5}$ cm **E** $3\sqrt{6}$ cm

11. Which one of the following is a simplification of

$$\frac{2}{x^2-1} - \frac{1}{x-1}$$

[where $x \neq \pm 1$]

A $-\frac{1}{x+1}$ **B** $\frac{2}{x-1}$ **C** $\frac{3-x}{(x-1)^2}$ **D** $\frac{1}{x^2-1}$ **E** $\frac{3-x}{(x-1)(x+1)}$

12. The straight-line graph given by the equation

$$\frac{x}{4} + \frac{y}{6} = 1$$

intersects the x-axis at $A\ (a, 0)$ and the y-axis at $B\ (0, b)$.

A circle passes through A and B and has a diameter AB.

What are the coordinates of the centre of the circle?

A (2, 3) **B** (3, 2) **C** (6, 4) **D** (4, 6) **E** (0, 0)

13. $x = 3 \times 10^m$ and $y = 5 \times 10^n$ where m and n are integers.

Which of the following is an expression, in scientific notation, for xy?

A $8 \times 10^{m+n}$ **B** 1.5×10^{mn} **C** $15 \times 10^{m+n-1}$ **D** $1.5 \times 10^{mn+1}$ **E** $1.5 \times 10^{m+n+1}$

14. David has two boxes containing shapes. In box A there are 4 stars and 2 hearts. In box B there are 2 stars and 1 heart. David takes, at random, a shape from box A and puts it into box B. He then takes a shape from box B.

What is the probability that this shape is a star?

A $\frac{1}{12}$ **B** $\frac{4}{9}$ **C** $\frac{2}{3}$ **D** $\frac{3}{4}$ **E** $\frac{4}{3}$

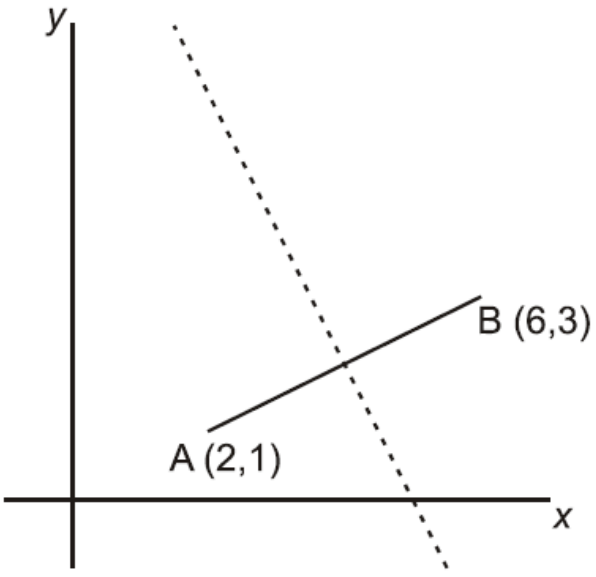
15. Which of the expressions below has the largest value for $0 < x < 1$?

A $\frac{1}{x}$ **B** x^2 **C** $\frac{1}{1+x}$ **D** $\frac{1}{\sqrt{x}}$ **E** $\sqrt{x}$

16. How many different integers, n, are there such that the difference between $2\sqrt{n}$ and 7 is less than 1?

A 0 **B** 2 **C** 4 **D** 6 **E** 8

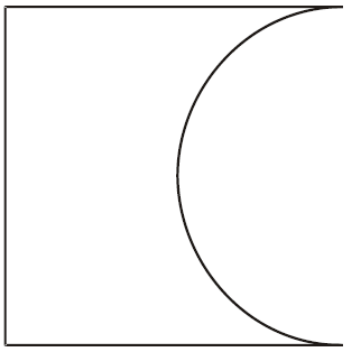
17. The graph below shows the line joining $A\ (2,1)$ and $B\ (6,3)$, and its perpendicular bisector (shown dashed: - - - - - - - - - -).



Which of the following is the equation of the dashed line?

A $y = 2x - 10$ **B** $y = 2x - 6$ **C** $y = 10 - 2x$ **D** $y = 4 - \frac{x}{2}$ **E** $y = 4 - 2x$

18. A square piece of metal has a semicircular piece cut out of it as shown. The area of the remaining metal is $100\ cm^2$.



Which of the following is a correct expression of the length of the side of the square in cm?

A $10\sqrt{\frac{1}{8-\pi}}$ **B** $10\sqrt{\frac{2}{4-\pi}}$ **C** $20\sqrt{\frac{2}{8+\pi}}$ **D** $20\sqrt{\frac{2}{8-\pi}}$ **E** $20\sqrt{\frac{1}{4-\pi}}$

19. Half the difference between two numbers is 2. The sum of the greater number and twice the smaller number is 13.

What is the smaller number?

A 2 **B** 3 **C** 4 **D** 5 **E** 6

20. A brother and a sister share out their collection of 5000 stamps in the ratio 5:3. The brother then shares his stamps with two friends in the ratio 3:1:1, keeping most for himself.

How many stamps do each of his friends receive?

A 525 **B** 585 **C** 625 **D** 825 **E** 1250

21. Given:

$$5n = \frac{m(ny - e^2)}{p} + n$$

Write the equation making y the subject.

A $y = \frac{3np+me^2}{mn}$ **B** $y = \frac{4np+me^2}{mn}$ **C** $y = \frac{3np+mne^2}{m}$ **D** $y = \frac{4nmp+me^2}{mn}$ **E** $y = 3n\left(\frac{p+me^2}{mn}\right)$

22. Given that p and q are non-zero vectors that are not parallel and that

$$3p + (x - 2)q = 2yp - 5q$$

Find the values of x and y.

A $x = 3, y = 1.5$ **B** $x = -3, y = 1.5$ **C** $x = 3, y = -1.5$ **D** $x = -3, y = -1.5$ **E** $x = 1.5, y = 3$

23. A wall is covered by 600 tiles which are $10\ cm$ by $10\ cm$.

How many $20\ cm$ by $20\ cm$ tiles are needed to cover the same floor?

A 100 **B** 120 **C** 140 **D** 150 **E** 160

24. Solve the inequalities

$$x + 2 \leq 5 + 3x \leq 2x + 7$$

A $-1.5 \leq x \leq -2$ **B** $-1.5 \leq x \leq 2$ **C** $1.5 \leq x \leq 2$ **D** $1.5 \leq x \leq -2$ **E** $-2.5 \leq x \leq 3$

25. A man and his wife are looking to buy a crate of lager.

The man and his wife agree to split the cost, in the ratio 3:2.

If the cost of the lager is doubled, in what ratio should they divide it so that the man still pays the same amount?

A 2:7 **B** 3:7 **C** 4:7 **D** 5:7 **E** 6:7

26. $£400$ is divided between Ann, Brian and Carol so that Ann has twice as much as Brian and Brian has three times as much as Carol.

How much does Brian receive?

A $£80$ **B** $£90$ **C** $£100$ **D** $£110$ **E** $£120$

27. A cake weighing $550\ g$ has three ingredients: flour, sugar and raisins. There is twice as much flour as sugar and 1.5 times as much sugar as raisins.

How much flour is there?

A $300\ g$ **B** $350\ g$ **C** $200\ g$ **D** $250\ g$ **E** $275\ g$

28. Given that $T = \frac{kx}{y}$

Find the percentage increase in T when k, x and y all increase by 20%

A 0.2% **B** 2% **C** 20% **D** 22% **E** 40%

29. Two men running towards each other at $4m/s$ and $6m/s$ respectively are one kilometer apart.

How long will it take before they meet?

A $80s$ **B** $90s$ **C** $100s$ **D** $110s$ **E** $120s$

30. David drove at an average speed of $50\ km/h$ for a distance of $45\ km$ in his car. David's journey took 54 minutes.

On a different journey, Marian drove at an average speed of $70\ km/h$. The time she took on her journey was 80% more than the time taken by David on his journey.

What is the distance, in km, that Marian drove?

A $56.5\ km$ **B** $113\ km$ **C** $226\ km$ **D** $56\ km$ **E** $85\ km$

Answer key:

№	Correct Answer	Your answer
1	D	
2	C	
3	E	
4	C	
5	B	
6	A	
7	D	
8	B	
9	A	
10	C	
11	A	
12	A	
13	E	
14	C	
15	A	
16	D	
17	C	
18	D	
19	B	
20	C	
21	B	
22	B	
23	D	
24	B	
25	B	
26	E	
27	A	
28	C	
29	C	
30	B	
		Total: _____